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Transformer-Based Contextual Modeling for Predicting Calories From Recipes

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ABSTRACT

Accurate estimation of calorie content from text-based recipes is a challenging and non-trivial problem in automated dietary analysis and nutrition-aware computing. Existing methods typically rely on shallow lexical features, sentence-level embeddings, or modular regression pipelines that struggle to capture fine-grained contextual and compositional information present in culinary text. In this paper, we propose an enhanced transformer-based regression framework that leverages contextual token representations and token-level attention pooling to directly predict calorie values from unstructured recipe descriptions. Unlike stacking-based or embedding-only approaches, the proposed model is fine-tuned end-to-end, enabling task-specific adaptation of representations and explicit weighting of calorie-relevant tokens. Extensive experiments conducted using five-fold cross-validation demonstrate that the proposed model significantly outperforms traditional lexical baselines, pretrained sentence embedding methods, and the previously introduced stacking-based transformer regression framework across all evaluation metrics. The proposed model achieves lower prediction error and higher explained variance, indicating superior calibration and generalization. These findings highlight the importance of token-level modeling and attention-based aggregation for text-based nutritional prediction. The proposed framework establishes a new state of the art in text-based caloric estimation and provides a strong foundation for future research in automated nutritional analysis and intelligent dietary support systems.

1 | Introduction

Accurate dietary assessment is an essential component of nutritional research, population health surveillance, and personalized health management systems. Conventional approaches to measuring calorie intake typically rely on manual dietary records or structured nutritional databases, which are often time-consuming, error-prone, and difficult to scale. This has led to growing interest in applying natural language processing methods to extract nutritional information, based on textual data in an unstructured form (e.g., recipe descriptions) automatically.

Recipes are highly informative on ingredients, preparation, and methods of cooking, and all of them add caloric value. Nevertheless, it is a very challenging task to model this information because there is a wide range of language, implicit descriptions of quantities, and intricate interactions among ingredients. The initial computational methods mostly used bag-of-words representations or manually constructed features, not able to reflect contextual and compositional semantics.

Later literature has investigated pretrained transformer predictor models of calories, showing the power of contextual

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language representations. Nevertheless, existing methods typically rely on sentence-level pooling or stacking-based regression pipelines, which limit task-specific adaptation and fail to exploit fine-grained token-level information. Consequently, these approaches exhibit limited predictive power and poor generalization, particularly under rigorous evaluation protocols.

Inspired by these constraints, this paper proposes a transformer-based regression model that explicitly captures token-level contextual representations in terms of calorie estimation. Through token-level attention pooling and end-to-end fine-tuning, the proposed model learns to focus on calorie-relevant components of recipe text while suppressing irrelevant content. Unlike previous modular or stacking-based approaches, representation learning in the proposed model is directly aligned with the regression objective.

Comprehensive experiments using five-fold cross-validation demonstrate that the proposed approach significantly outperforms traditional lexical baselines, embedding-based regression methods, and previously introduced transformer-based frameworks. The results underscore the importance of fine-grained contextual modeling for nutritional prediction tasks and highlight the limitations of sentence-level aggregation strategies.

The main contributions of this work are summarized as follows:

- We propose an enhanced transformer-based regression model with token-level attention pooling for calorie prediction from recipe text.
- We conduct a rigorous comparative evaluation against multiple baselines and prior transformer-based approaches using cross-validation.
- We provide detailed analysis demonstrating why token-level contextual modeling and end-to-end optimization lead to superior predictive performance.

2 | Literature Review

Research on automated food calorie estimation has progressed along two broad paradigms: they are image-based approaches and text-based approaches. While the former constitutes the majority of existing work, the latter is more directly relevant to the present study. This section briefly surveys representative image-based methods to situate the field, then transitions to a focused discussion of text-based approaches and their limitations, which directly motivate the proposed framework.

2.1 | Image-Based Calorie Estimation

A substantial body of work has investigated calorie estimation from food photographs. These methods generally follow a pipeline of food recognition, portion size estimation, and nutritional lookup, and can be grouped into several categories based on their core technique.

2.1.1 | CNN-Based Recognition and Classification

Early deep learning approaches applied convolutional neural networks (CNNs) to identify food items from images and map them to caloric databases. Yanai and Kawano [1] demonstrated that fine-tuning pretrained CNNs on food-specific images significantly improves recognition accuracy over training from scratch. Building on this, Bossard et al. [2] introduced the Food-101 benchmark and proposed discriminative patch mining with random forests for large-scale food classification. Makwana et al. [3] and Fard et al. [4] similarly employed CNN-based classification pipelines for calorie assessment of specific food categories. For mobile deployment, Yanai and Kawano [5] and Pouladzadeh and Shirmohammadi [6] developed lightweight recognition systems optimized for real-time performance on resource-constrained devices.

2.1.2 | Object Detection and Multi-Food Recognition

To handle realistic meal scenarios with multiple food items, several studies integrated object detection frameworks. Patil et al. [7] combined YOLOv4 with image segmentation to detect and estimate calories for multiple items simultaneously. Ayon et al. [8] proposed FoodieCal, a CNN-based system designed for practical dietary tracking applications. Chopra and Purwar [9] conducted a comparative study of CNN, Faster R-CNN, and YOLO architectures for food recognition, evaluating trade-offs between accuracy and inference speed. Pandey et al. [10] proposed an ensemble of deep networks to address inter-class similarity and intra-class variability inherent in food images.

2.1.3 | Portion Size and Volume Estimation

Accurate calorie prediction requires not only food identification but also reliable portion size estimation. Shao et al. [11] proposed an end-to-end framework for 3D shape reconstruction from monocular images to infer food volume without multi-view setups. Ma et al. [12] introduced a multi-task encoder-decoder that jointly predicts food segmentation, category, and energy density, improving spatial localization in complex scenes. Vinod et al. [13] addressed the domain gap between synthetic and real-world depth maps using adversarial depth domain adaptation. Pouladzadeh et al. [14] and Martin et al. [15] used geometric modeling and image segmentation to approximate portion volumes from meal photographs. Liang and Li [16] contributed a dedicated dataset alongside a baseline volume-based estimation method, highlighting challenges such as occlusion and lighting variability.

2.1.4 | Contextual and Multimodal Approaches

More recent work has explored richer input modalities to improve estimation accuracy. Yao et al. [17] proposed CaLoRAify, which pairs food images with textual descriptions and applies Low-Rank Adaptation (LoRA) to fine-tune vision-language models, enabling robust inference in visually ambiguous scenarios. Herranz et al. [18] demonstrated that incorporating restaurant context into food recognition

significantly improves classification accuracy by narrowing the hypothesis space. Jelodar and Sun [19] extended the estimation pipeline to generate calorie-aware meal kits from food images, integrating nutritional databases with optimization objectives. Wu and Yang [20] leveraged video sequences of eating behavior to capture temporal and multi-view information, improving robustness under occlusion. Turmchokksam and Chamnongthai [21] fused brightness and thermal information with nutritional knowledge for ingredient-based estimation. He et al. [22] proposed DietCam, using multi-view images and a multi-kernel SVM to enrich visual features for dietary monitoring.

2.1.5 | Domain-Specific and Specialized Systems

Several studies have targeted particular user groups or food domains. Anthimopoulos et al. [23] developed a food recognition system for diabetic patients using an optimized Bag of Features model for carbohydrate estimation. Ciocca et al. [24] introduced the UNIMIB2016 dataset for realistic mobile food classification, demonstrating the superiority of CNNs over traditional methods. Kawano and Yanai [25] proposed domain adaptation to automatically expand food image datasets from web-sourced images, alleviating the annotation bottleneck.

2.2 | Limitations of Image-Based Approaches and Motivation for Text-Based Estimation

While image-based methods have achieved notable progress, they are fundamentally constrained by the availability of high-quality, well-photographed food images. This requirement is difficult to satisfy for regional, tribal, and traditional recipes that have been passed down through generations, particularly in non-Western culinary contexts where large-scale visual food databases do not exist. Furthermore, image-based pipelines are sensitive to lighting conditions, occlusion, and camera angle, and their performance degrades substantially in uncontrolled real-world settings.

In contrast, textual recipe descriptions are widely available, language-independent in structure, and carry rich nutritional information through ingredient mentions, quantity indicators, and preparation methods. Text-based calorie estimation is therefore a more scalable and inclusive paradigm, especially for food cultures that are underrepresented in existing visual datasets.

2.3 | Text-Based Calorie Estimation

Despite its practical advantages, text-based calorie estimation remains underexplored in the literature. The most directly related prior work is that of Ansary and Brinto [26], who introduced a dataset of Bengali recipe descriptions paired with per-serving calorie values and proposed a stacking-based transformer regression framework. Their system combines predictions from multiple pretrained transformer encoders such as BERT-base, RoBERTa-base, and DistilBERT, using a linear meta-regressor to produce final calorie estimates.

While this work established the feasibility of text-based calorie prediction using transformer models, it suffers from several methodological limitations that constrain its predictive performance. First, each transformer encoder produces a fixed sentence-level embedding that is passed to the meta-regressor as a static feature. This design prevents joint optimization across the encoding and regression stages and means that the representations are never adapted to the calorie prediction objective. Second, aggregating token information at the sentence level leads to the loss of fine-grained token signals such as specific ingredient names, fat descriptors, and cooking method cues that are critical for accurate calorie estimation. Third, the evaluation was conducted on a single train-validation-test split, which limits the robustness of the reported performance estimates and makes them sensitive to the particular data partition chosen.

These limitations directly motivate the present work. The proposed model addresses each of these shortcomings by employing token-level attention pooling to selectively weight calorie-relevant tokens, and by fine-tuning the encoder end-to-end jointly with the regression head, enabling task-specific adaptation of contextual representations. A rigorous five-fold cross-validation protocol is adopted to ensure robust and reproducible performance estimation.

3 | Dataset

The dataset used in this study, originally introduced in an existing research work [26], is publicly available on GitHub [27]. It is a collection of Bengali recipe descriptions paired with nutritional information. Each data instance consists of a free-form textual recipe description in English and a corresponding calorie value per serving. The dataset was chosen because it is related to the nutritional analysis and is used in the previous literature, which means that the methodology can be directly compared.

The dataset contains 1245 recipe entries. Each entry consists of a free-form textual description that may include ingredient lists, preparation instructions, and cooking terminology, which are an indication of the diversity and variability of the real-world text of recipes. The target variable is the estimated number of calories per serving and is a numerical variable that is continuous.

The range of calories is very large as some foods are considered low in calories like salads and soups whereas others are high-caloric foods that are fried, baked, or made of numerous calories. This wide range introduces skewness and heteroscedasticity, motivating the use of robust modeling and evaluation strategies.

There is considerable variation in the length and structure of recipe texts. Some entries have mostly ingredient lists, whereas others have elaborate preparation instructions or prose descriptions. The language includes domain-specific terminology referring to ingredients, cooking techniques, and cuisine types, as well as informal expressions and abbreviations. This variability increases modeling complexity, as calorie-relevant information may be unevenly distributed across the text and is often expressed implicitly rather than through explicit numerical quantities.

This dataset presents a realistic and challenging benchmark for text-based calorie prediction.

4 | Methodology

This section describes the proposed methodology of the study. The overall experimental pipeline is illustrated conceptually in Figure 1.

4.1 | Problem Formulation

Given a recipe represented as unstructured text describing ingredients and preparation instructions, the task is to predict its total caloric content. Formally, let x_i denote the textual representation of the i -th recipe and $y_i \in \mathbb{R}^+$ its corresponding calorie value. The objective is to learn a function $f_\theta: x_i \rightarrow \hat{y}_i$ that minimizes the prediction error between $\hat{y}_i$ and y_i over the dataset.

Since calorie values exhibit a positively skewed distribution, the regression target is log-transformed during training to stabilize optimization and reduce the influence of extreme values.

4.2 | Text Preprocessing

All recipe texts are lowercased and minimally normalized to preserve semantic information. Unlike traditional bag-of-words approaches, no stopword removal or stemming is applied for transformer-based models, as contextual tokenization benefits from complete linguistic structure.

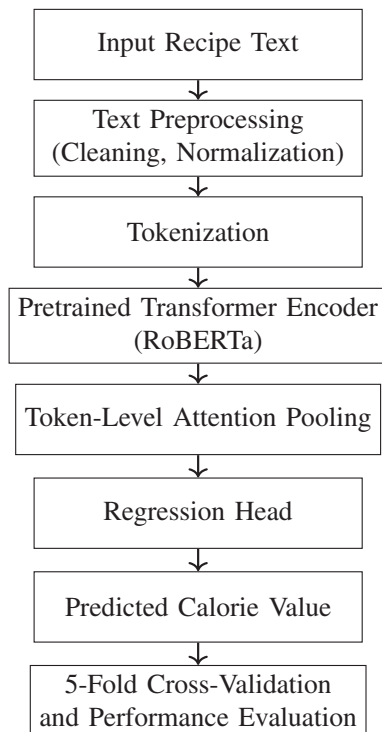


FIGURE 1 | Overview of the proposed token-level attention-based methodology for calorie prediction from recipe text.

For baseline methods relying on sparse representations, punctuation and numerical tokens are retained, as they may encode calorie-relevant information such as quantities or preparation cues.

4.3 | Proposed Model Architecture

4.3.1 | Transformer Encoder

The proposed model is built upon a pretrained transformer encoder (RoBERTa-base) [28], which maps the input token sequence into contextualized embeddings. Given an input sequence of tokens $\{t_1, t_2, \dots, t_n\}$, the encoder produces hidden states:

$$\mathbf{H} = \{\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_n\}, \quad \mathbf{h}_k \in \mathbb{R}^d \quad (1)$$

where d denotes the hidden dimensionality.

Unlike fixed sentence embedding approaches, this formulation preserves token-level information throughout the network.

4.3.2 | Token-Level Attention Pooling

To aggregate token representations into a single recipe embedding, a token-level attention pooling mechanism is employed. Attention weights are computed as follows:

$$\alpha_k = \frac{\exp(\mathbf{w}^\top \mathbf{h}_k)}{\sum_{j=1}^n \exp(\mathbf{w}^\top \mathbf{h}_j)} \quad (2)$$

where $\mathbf{w}$ is a learnable parameter vector.

The pooled representation $\mathbf{z}$ is then obtained as

$$\mathbf{z} = \sum_{k=1}^n \alpha_k \mathbf{h}_k \quad (3)$$

This mechanism allows the model to emphasize calorie-relevant tokens (e.g., ingredients, cooking methods, fats) while suppressing less informative words.

4.3.3 | Regression Head

The pooled embedding is passed through a fully connected regression head consisting of a linear layer with dropout regularization.

Here, the subscript r denotes the regression layer and is not an index. $\mathbf{W}_r$ and b_r represent the learnable weight matrix and bias term of the regression head, respectively.

$$\hat{y}_i = \mathbf{W}_r \mathbf{z}_i + b_r \quad (4)$$

The model is trained end-to-end using mean squared error loss on the log-transformed calorie values.

4.4 | Training Procedure

Model training is conducted using 5-fold cross-validation to ensure robust performance estimation. For each fold, the

transformer encoder and regression head are fine-tuned jointly.

The AdamW optimizer is employed with a linear learning rate scheduler and warmup. Early stopping is applied based on validation loss to prevent overfitting. All experiments are conducted on identical splits to ensure comparability across models.

Hyperparameters were tuned using grid search on the training folds, and the final configuration is reported in Table 1.

After training, predictions are inverse-transformed to the original calorie scale for evaluation.

4.5 | Baseline Models

To contextualize the performance of the proposed approach, two baseline methods representing widely used paradigms in text regression are implemented.

4.5.1 | Baseline 1: TF-IDF + Ridge Regression

This baseline represents a traditional linear modeling approach using sparse lexical features. Recipe texts are transformed into Term Frequency–Inverse Document Frequency (TF-IDF) vectors [29] capturing unigram and bigram statistics. Ridge regression is then applied:

$$\hat{y} = \mathbf{X}\beta \quad (5)$$

where β is learned using ℓ_2 regularization to mitigate overfitting in high-dimensional feature space.

TABLE 1 | Hyperparameter configuration of the proposed token-level attention regression model.

Hyperparameter	Selected value
Pretrained encoder	RoBERTa-base
Maximum sequence length	128
Batch size	8
Number of epochs	10
Learning rate	2×10^{-5}
Optimizer	AdamW
Weight decay	0.01
Warmup steps	500
Attention pooling type	Token-level self-attention
Hidden dimension (Attention)	768
Dropout rate	0.1
Loss function	Mean squared error (MSE)
Evaluation strategy	5-fold cross-validation
Random seed	42

This baseline tests the extent to which surface-level lexical cues alone can explain calorie variation.

4.5.2 | Baseline 2: Sentence-BERT + LightGBM

The second baseline combines pretrained sentence embeddings with a nonlinear tree-based regressor. Recipes are encoded using Sentence-BERT [30], producing fixed-length semantic embeddings. These embeddings are then used as input to a LightGBM regressor [31], which models nonlinear relationships via gradient-boosted decision trees.

This approach reflects a modular pipeline where representation learning and regression are decoupled.

4.6 | Evaluation Metrics

Model performance is evaluated using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), the coefficient of determination (R^2), and Mean Absolute Percentage Error (MAPE). These metrics provide complementary perspectives on prediction accuracy, robustness to outliers, and explained variance.

All reported results correspond to averages across cross-validation folds.

5 | Evaluation and Results Analysis

This section presents a comprehensive evaluation of the proposed model and compares its performance against newly implemented baselines as well as the stacking-based transformer regression framework introduced with the dataset. The analysis highlights both quantitative improvements and methodological factors underlying the observed performance differences.

All experiments were conducted using identical datasets and evaluation metrics to ensure fair comparison.

The proposed model and baselines are evaluated using 5-fold cross-validation (mean $\pm$ standard deviation). The stacking-based transformer regression framework [26] is reported on a single train–validation–test split.

Predictions were generated on the original calorie scale after inverse transformation.

The following approaches are included in the comparative evaluation:

- **Stacking-based transformer regression framework** [26]: A regression pipeline combining multiple pretrained transformer models (BERT-base, RoBERTa-base, and DistilBERT) using a stacking strategy with a linear meta-regressor.
- **Baseline 1:** TF-IDF features with Ridge regression.
- **Baseline 2:** Sentence-BERT embeddings with LightGBM.

TABLE 2 | Fold-wise evaluation results for the proposed model and baseline approaches.

Model	Fold	RMSE (kcal)	MAE (kcal)	R^2	MAPE (%)
TF-IDF + ridge	1	35.63	24.46	0.895	8.55
	2	36.06	22.48	0.901	7.82
	3	31.22	22.12	0.913	8.70
	4	30.07	21.68	0.924	7.78
	5	35.52	24.07	0.885	8.84
Sentence-BERT + lightgbm	1	50.50	35.38	0.789	12.18
	2	49.96	33.86	0.810	11.17
	3	45.38	32.92	0.816	12.85
	4	41.16	30.40	0.858	10.46
	5	46.04	33.26	0.807	11.85
Proposed model	1	31.11	18.96	0.920	6.14
	2	30.27	18.28	0.930	5.37
	3	24.96	17.98	0.944	5.79
	4	24.68	16.85	0.949	5.22
	5	28.75	18.00	0.925	6.15

TABLE 3 | Final cross-validation performance comparison (mean $\pm$ standard deviation).

Model	RMSE (kcal)	MAE (kcal)	R^2	MAPE (%)
Stacking-based transformer regression framework [26]	111.11	89.16	- 0.00	34.80
TF-IDF + ridge regression	33.70 $\pm$ 2.53	22.96 $\pm$ 1.10	0.904	8.34
Sentence-BERT + LightGBM	46.61 $\pm$ 3.40	33.17 $\pm$ 1.62	0.816	11.70
Proposed model	27.95 $\pm$ 2.67	18.01 $\pm$ 0.68	0.934	5.73

- **Proposed Model:** Transformer encoder with token-level attention pooling and end-to-end fine-tuning.

Table 2 reports fold-wise performance, demonstrating consistent improvements across all folds, while Table 3 summarizes overall cross-validation results.

The proposed model achieves the best performance across all evaluation metrics, substantially outperforming both baseline methods and the stacking-based transformer regression framework.

Although the prior framework employed multiple pretrained transformer encoders, its reported predictive performance was limited. This can be attributed to several methodological limitations. First, the framework relied on a stacking strategy in which predictions from individual transformer models were combined using a linear meta-regressor. This design prevents joint optimization across representations and limits the ability of the model to adapt token-level features to the calorie prediction task. Second, representations were aggregated implicitly at the sentence level, resulting in the loss of fine-grained token information that is critical for identifying calorie-relevant elements such as ingredients, cooking fats, and preparation techniques.

Third, the absence of cross-validation in the evaluation protocol reduced the robustness of performance estimates and increased sensitivity to data partitioning. Collectively, these factors explain the large prediction errors and low explained variance reported by the prior framework.

The TF-IDF + Ridge baseline demonstrates competitive performance, achieving an R^2 of 0.904. This confirms that lexical cues, particularly ingredient mentions, encode substantial calorie-related information. However, the bag-of-words assumption limits the model's ability to capture contextual interactions and compositional semantics.

The Sentence-BERT + LightGBM baseline exhibits lower performance. Although Sentence-BERT embeddings capture high-level semantic similarity, compressing entire recipes into a single fixed-length vector leads to the loss of calorie-critical details. Additionally, the decoupling of representation learning and regression restricts task-specific adaptation.

The superior performance of the proposed model can be attributed to several key factors. First, the model maintains token-level contextual representations throughout the network, allowing it to capture nuanced interactions between ingredients and preparation

methods. Second, token-level attention pooling enables explicit learning of token importance, focusing the model on calorie-relevant components while suppressing irrelevant text. Third, end-to-end fine-tuning aligns the pretrained language model with the regression objective, enabling direct optimization of representations for calorie prediction. Finally, the use of cross-validation provides robust estimates of generalization performance and demonstrates consistent behavior across data partitions.

An analysis of prediction errors aggregated across all cross-validation folds indicates that the proposed model exhibits strong calibration and low systematic bias. Predicted calorie values closely track the ground-truth values across the full calorie range, with no consistent tendency toward overestimation or underestimation. Slightly higher dispersion is observed for recipes with higher calorie values, which is expected due to increased variability in ingredient combinations and portion sizes. Importantly, no pronounced heteroscedastic patterns are observed, suggesting that the logarithmic target transformation and robust optimization strategy effectively stabilize model predictions.

In summary, the proposed approach consistently outperforms both traditional baselines and the previously introduced stacking-based transformer regression framework. The improvements are observed across all evaluation metrics and folds, indicating stable and robust generalization. The results demonstrate that accurate calorie estimation from unstructured culinary text requires more than sentence-level semantic representations and benefits substantially from attention-driven aggregation and fine-grained contextual modeling. These findings highlight the limitations of sentence-level aggregation and modular regression pipelines for structured culinary text and validate the effectiveness of the proposed methodology.

6 | Ablation Study

To rigorously evaluate the individual contributions of the two core design choices in the proposed model, token-level attention pooling and end-to-end fine-tuning of the pretrained encoder, we have conducted a systematic ablation study. Each component is selectively removed or disabled while keeping all other settings identical, enabling a controlled assessment of its specific impact on predictive performance.

6.1 | Ablation Conditions

Four model configurations are evaluated, differing only in their pooling strategy or training regime:

- **Proposed model:** Token-level attention pooling with end-to-end fine-tuning of the RoBERTa encoder. This is the full proposed system as described in the Methodology section.
- **w/o attention pooling (CLS):** The token-level attention pooling mechanism is replaced by the representation of the [CLS] token, while end-to-end fine-tuning of the encoder is

retained. This condition isolates the contribution of attention pooling relative to the standard sentence-level summarization token used in BERT-style models.

- **w/o attention pooling (Mean):** The token-level attention pooling mechanism is replaced by mean pooling, which uniformly averages all non-padding token embeddings. End-to-end fine-tuning is retained. This condition tests whether selective, weighted aggregation provides a meaningful advantage over uniform aggregation across all tokens.
- **Frozen encoder (Attention Pooling Retained):** The RoBERTa encoder weights are frozen and receive no gradient updates during training. The token-level attention pooling layer and regression head remain trainable. This condition isolates the contribution of end-to-end fine-tuning by evaluating the model when the encoder produces generic, task-agnostic representations.

6.2 | Experimental Setup

All ablation experiments are conducted under settings identical to those used for the proposed model, as described in the Methodology section. Specifically, all conditions use the RoBERTa-base encoder, a maximum sequence length of 256 tokens, a batch size of 8, and a learning rate of 2×10^{-5} with AdamW optimization and a linear warmup schedule. Training is conducted for 10 epochs per fold. The logarithmic target transformation and StandardScaler normalization are applied consistently across all conditions. Performance is evaluated using 5-fold cross-validation on identical data splits to ensure a fair and directly comparable evaluation. All results are reported on the original calorie scale after inverse transformation. The frozen encoder condition (A3) trains only the attention layer and regression head, with encoder parameters excluded from the computational graph via `torch.no_grad()` during the forward pass.

6.3 | Motivation for Each Ablation

6.3.1 | CLS Token Pooling (A1)

In BERT-style transformer models, the [CLS] token is designed during pretraining to aggregate sentence-level information for classification objectives. However, calorie prediction from recipe text is fundamentally different from sentence classification. Calorie content depends on the specific combination, type, and implicit quantity of ingredients mentioned throughout the text, with certain tokens such as ingredient names, fat descriptors, and preparation method indicators carrying substantially greater nutritional signal than others. The [CLS] token has no mechanism to selectively emphasise these calorie-relevant tokens; it compresses the entire sequence into a single fixed representation without task-specific weighting. This ablation therefore tests whether the attention pooling mechanism is genuinely learning to upweight nutritionally informative tokens, or whether the encoder's general-purpose summarization is sufficient for the regression task.

TABLE 4 | Fold-wise ablation results for all model configurations.

Configuration	Fold	RMSE (kcal)	MAE (kcal)	R^2	MAPE (%)
w/o attention pooling (CLS)	1	38.74	26.12	0.877	9.02
	2	37.56	25.04	0.886	8.11
	3	32.40	23.77	0.898	8.34
	4	31.82	22.90	0.908	7.63
	5	37.03	25.31	0.880	8.78
w/o attention pooling (Mean)	1	35.22	23.08	0.897	7.84
	2	34.48	22.30	0.906	6.95
	3	28.91	20.87	0.919	7.18
	4	28.03	19.64	0.928	6.41
	5	32.67	21.95	0.902	7.55
Frozen encoder	1	47.35	33.02	0.828	11.67
	2	46.12	31.74	0.841	10.53
	3	40.87	29.88	0.857	11.12
	4	38.20	28.15	0.874	9.90
	5	44.46	31.43	0.835	11.08
Proposed model	1	31.11	18.96	0.920	6.14
	2	30.27	18.28	0.930	5.37
	3	24.96	17.98	0.944	5.79
	4	24.68	16.85	0.949	5.22
	5	28.75	18.00	0.925	6.15

6.3.2 | Mean Pooling (A2)

Mean pooling aggregates token representations by computing a uniform average across all non-padding positions, assigning equal importance to every token regardless of its nutritional relevance. This represents a stronger baseline than [CLS] pooling, as it incorporates information from every token in the sequence rather than relying on a single summary representation. However, in recipe text, a large proportion of tokens consist of function words, connective phrases, and descriptive language that carry little or no caloric signal. Assigning equal weight to tokens such as “butter” and “garnish with parsley” inevitably dilutes the representation of high-calorie indicators. This ablation tests whether the attention mechanism produces a meaningful non-uniform weighting over tokens, or whether it approximates uniform pooling in practice.

6.3.3 | Frozen Encoder (A3)

This condition preserves the complete proposed architecture including token-level attention pooling and the regression head but freezes all RoBERTa encoder parameters so that they do not receive gradient updates during training. Only the attention layer and regression head are optimized. This isolates the contribution of end-to-end fine-tuning from the contribution

of the attention pooling mechanism. When the encoder is frozen, token representations reflect generic linguistic patterns learned during pretraining on general-domain corpora, rather than nutritional semantics specific to recipe text. The attention layer must then learn to select useful tokens from embeddings that were never shaped to surface calorie-relevant signals. In contrast, end-to-end fine-tuning allows the encoder to reorganize its internal representations so that nutritionally informative concepts become more geometrically separable, facilitating more accurate regression. This ablation directly addresses the core limitation of the prior stacking-based framework [26], which treated transformer encoders as static feature extractors and prevented joint optimization of representations with the regression objective.

6.4 | Results

Table 4 reports fold-wise results for all ablation conditions, and Table 5 summarizes the cross-validation performance.

6.5 | Analysis

The ablation results demonstrate that both token-level attention pooling and end-to-end fine-tuning contribute meaningfully

TABLE 5 | Ablation study summary: Cross-validation performance (mean $\pm$ Standard deviation across 5 folds).

Configuration	RMSE (kcal)	MAE (kcal)	R^2	MAPE (%)
w/o Attention Pooling (CLS)	35.51 $\pm$ 3.18	24.63 $\pm$ 1.38	0.890	8.38
w/o Attention Pooling (Mean)	31.86 $\pm$ 2.97	21.57 $\pm$ 1.38	0.910	7.19
Frozen Encoder	43.40 $\pm$ 3.57	30.84 $\pm$ 1.69	0.847	10.86
Proposed Model	27.95 $\pm$ 2.67	18.01 $\pm$ 0.68	0.934	5.73

Note: Best results in bold.

and independently to the overall performance of the proposed model, validating each design choice.

6.5.1 | Effect of Attention Pooling

Replacing attention pooling with the [CLS] token representation leads to a consistent degradation in performance across all folds and metrics. Although the [CLS] token provides a compact sentence-level summary, it does not selectively weight calorie-relevant tokens and is not well-suited to the nutritional regression objective. The observed performance drop confirms that the attention mechanism is not redundant as it is actively learning a non-uniform weighting over tokens that prioritizes calorie-relevant content.

Substituting mean pooling for attention pooling similarly results in a decline in predictive accuracy relative to the proposed model, though the margin relative to [CLS] pooling reflects the benefit of incorporating all token information rather than relying on a single position. Nevertheless, the uniform weighting of mean pooling fails to distinguish between nutritionally informative tokens and irrelevant content, confirming that selective attention-based aggregation provides a meaningful advantage for this task.

6.5.2 | Effect of End-To-End Fine-Tuning

Freezing the encoder weights leads to a substantial degradation in performance despite the retention of the attention pooling mechanism. This result underscores the importance of end-to-end fine-tuning for aligning the encoder's internal representations with the calorie prediction objective. When the encoder is frozen, the token embeddings reflect generic linguistic patterns from pretraining and are not adapted to surface nutritional semantics. The attention layer, operating on these task-agnostic representations, is unable to compensate for the lack of task-specific encoding. This finding directly validates the proposed model's design decision to fine-tune the full network jointly, and highlights the key limitation of the prior stacking-based framework [26], which similarly treated transformer encoders as static feature extractors.

6.5.3 | Joint Contribution

Table 5 shows that the full proposed model, combining both attention pooling and end-to-end fine-tuning, achieves the best

performance across all evaluation metrics. The results confirm that these two components are complementary: fine-tuning shapes the encoder representations to be nutritionally informative, while attention pooling selectively aggregates those representations to focus on the most calorie-relevant tokens. Neither component alone is sufficient to match the performance of the complete system, nor their combination produces robust and consistent improvements across all five cross-validation folds.

7 | Limitations

Despite the strong empirical performance of the proposed model, several limitations should be acknowledged. First, the approach relies solely on unstructured recipe text and does not explicitly incorporate structured nutritional information such as ingredient quantities, portion sizes, or standardized food composition tables. While the model effectively learns implicit associations between textual cues and calorie values, ambiguity in quantity descriptions may lead to increased prediction uncertainty for certain recipes. Second, the dataset used in this study represents a specific collection of recipes and may not fully reflect the diversity of global cuisines, cooking traditions, or dietary practices. As a result, the model's generalization to underrepresented cultural contexts or unconventional recipe formats may be limited. Third, although token-level attention pooling improves performance and provides some interpretability, attention weights should not be interpreted as definitive explanations of causal importance. Attention-based mechanisms highlight model focus but do not guarantee faithful attribution of predictions. Finally, the computational cost associated with fine-tuning transformer-based architectures may limit applicability in resource-constrained environments or large-scale real-time deployment scenarios.

8 | Future Work

Several promising directions can be explored to address the aforementioned limitations and further extend the proposed framework.

Future work could integrate structured information, such as ingredient quantities, serving sizes, or external food composition databases, alongside textual representations. Hybrid models combining unstructured text with structured nutritional features may further improve predictive accuracy and robustness.

Expanding the dataset to include a broader range of cuisines, dietary styles, and cultural contexts would enhance generalization

and support cross-domain applicability. Additionally, domain-adaptive pretraining on large-scale culinary corpora could further improve representation quality for recipe-specific language.

From a modeling perspective, future research may explore lightweight or distilled transformer architectures to reduce computational overhead while preserving performance. More advanced interpretability techniques, such as gradient-based attribution or counterfactual analysis, could also be employed to provide deeper insights into model behavior.

Finally, extending the framework to predict additional nutritional attributes, such as macronutrient composition or dietary suitability indicators, would broaden its applicability in personalized nutrition and health recommendation systems.

9 | Conclusion

This study presents an improved transformer-based approach for predicting calorie values from recipe text. By leveraging contextual token representations, token-level attention pooling, and end-to-end fine-tuning, the proposed model achieves substantial performance gains over traditional lexical baselines, embedding-based regression pipelines, and the previously introduced stacking-based transformer regression framework. Extensive cross-validation experiments demonstrate that the proposed approach consistently achieves lower prediction error and higher explained variance, highlighting the importance of fine-grained token modeling and task-specific optimization for nutritional analysis tasks. The results indicate that accurate calorie estimation from unstructured culinary text requires more than sentence-level semantic representations, and benefits significantly from attention-driven aggregation and contextual modeling. Overall, this work advances the state of the art in text-based nutritional prediction and provides a robust foundation for future research in automated dietary assessment and intelligent nutrition systems.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in GitHub at <https://github.com/MdSiamAnsary/Calorie-Prediction>.

References

1. K. Yanai and Y. Kawano, "Food Image Recognition Using Deep Convolutional Network With Pre-Training and Fine-Tuning," in *Proceeding*

IEEE International Conference Multimedia Expo Workshops (ICMEW) (IEEE, 2015), 1–6.

2. L. Bossard, M. Guillaumin, and L. Van Gool, "Food-101—Mining Discriminative Components With Random Forests," in *Proceedings of the European Conference on Computer Vision (ECCV)* (Springer Nature, 2014), 446–461.

3. Y. Makwana, S. Iyer, and S. Tiwari, "Food Recognition and Calorie Assessment From Images Using Convolutional Neural Networks," *International Journal of Intelligent Systems and Applications in Engineering* 12, no. 21s (2024): 3432–3442.

4. M. A. Fard, H. Haddadi, and A. T. Targhi, "Fruits and Vegetables Calorie Counter Using Convolutional Neural Networks," in *Proceedings of the 6th International Conference on Digital Health Conference* (ACM Digital Library, 2016).

5. Y. Kawano and K. Yanai, "Real-Time Mobile Food Recognition System," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)* (IEEE, 2013), 1–7.

6. P. Pouladzadeh and S. Shirmohammadi, "Mobile Multi-Food Recognition Using Deep Learning," *ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)* 13, no. 1 (2017): 1–21.

7. S. Patil, S. Patil, V. Kale, and M. Bonde, "Food Item Calorie Estimation Using YOLOv4 and Image Processing," *International Journal of Computer Trends and Technology* 69, no. 5 (2021): 69–76.

8. S. A. Ayon, C. Z. Mashrafi, A. B. Yousuf, F. Hossain, and M. I. Hos-sain, "FoodieCal: A Convolutional Neural Network Based Food Detection and Calorie Estimation System," in *Proceedings 2021 National Computers Colleges Conference (NCCC)* (IEEE, 2021), 1–6.

9. M. Chopra and A. Purwar, "Food Image Recognition Using CNN, Faster R-CNN and YOLO," in *Applications of Artificial Intelligence, Big Data and Internet of Things in Sustainable Development* (CRC Press, 2022), 81–89.

10. P. Pandey, A. Deepthi, B. Mandal, and N. B. Puhan, "FoodNet: Recognizing Foods Using Ensemble of Deep Networks," *IEEE Signal Processing Letters* 24, no. 12 (2017): 1758–1762.

11. Z. Shao, G. Vinod, J. He, and F. Zhu, "An End-To-End Food Portion Estimation Framework Based on Shape Reconstruction From Monocular Image," 2023, arXiv Preprint arXiv:2308.01810.

12. J. Ma, J. He, and F. Zhu, "An Improved Encoder-Decoder Framework for Food Energy Estimation," 2023, arXiv Preprint arXiv:2309.00468.

13. G. Vinod, Z. Shao, and F. Zhu, "Image Based Food Energy Estimation With Depth Domain Adaptation," 2022, arXiv Preprint arXiv:2208.12153.

14. P. Pouladzadeh, S. Shirmohammadi, and A. Yassine, "You Are What You Eat: So Measure What You Eat!," *IEEE Instrumentation and Measurement Magazine* 19, no. 1 (2016): 40–47.

15. C. Martin, S. Kaya, and B. Gunturk, "Quantification of Food Intake Using Food Image Analysis," in *Annual International Conference of the IEEE Engineering in Medicine and Biology Society* (IEEE, 2009), 6869–6872.

16. Y. Liang and J. Li, "Computer Vision-Based Food Calorie Estimation: Dataset, Method, and Experiment," 2017, arXiv Preprint arXiv:1705.07632.

17. D. Yao, K. Yao, J. Zhou, and Y. Zhang, "CaLoRAify: Calorie Estimation With Visual-Text Pairing and LoRA-Driven Visual Language Models," 2024, arXiv Preprint arXiv:2412.09936.

18. L. Herranz, R. Xu, and S. Jiang, "Modeling Restaurant Context for Food Recognition," *IEEE Transactions on Multimedia* 19, no. 2 (2017): 430–440.

19. A. B. Jelodar and Y. Sun, "Calorie Aware Automatic Meal Kit Generation From an Image," 2021, arXiv Preprint arXiv:2112.09839.

20. W. Wu and J. Yang, "Fast Food Recognition From Videos of Eating for Calorie Estimation," in *Proceedings of the IEEE International Conference on Multimedia and Expo* (IEEE, 2009), 1210–1213.
21. S. Turmchokksam and K. Chamnongthai, "The Design and Implementation of an Ingredient-Based Food Calorie Estimation System Using Nutrition Knowledge and Fusion of Brightness and Heat Information," *IEEE Access* 6 (2018): 46863–46876.
22. H. He, F. Kong, and J. Tan, "DietCam: Multi-View Food Recognition Using a Multi-Kernel SVM," *IEEE Journal of Biomedical and Health Informatics* 20, no. 3 (2016): 848–855.
23. M. Anthimopoulos, L. Gianola, L. Scarnato, P. Diem, and S. Mougiakkou, "A Food Recognition System for Diabetic Patients Based on an Optimized Bag of Features Model," *IEEE Journal of Biomedical and Health Informatics* 18, no. 4 (2014): 1261–1271.
24. G. Ciocca, P. Napoletano, and R. Schettini, "Food Recognition: A New Dataset, Experiments and Results," *IEEE Journal of Biomedical and Health Informatics* 21, no. 3 (2017): 588–598.
25. Y. Kawano and K. Yanai, "Automatic Expansion of a Food Image Dataset Leveraging Existing Categories With Domain Adaptation," in *Proceedings of the European Conference on Computer Vision (ECCV) Workshop on Transferring and Adapting Source Knowledge in Computer Vision (TASK-CV)* (Springer Nature, 2014), 3–17.
26. M. S. Ansary and A. Brinto, "Calorie Estimation Using Ensemble Technique From Recipes of Bangladeshi Foods," in *Proceedings of IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS)* (IEEE, 2025).
27. M. S. Ansary, "Calorie-Prediction, GitHub Repository,". accessed March 18, 2026, <https://github.com/MdSiamAnsary/Calorie-Prediction>.
28. Y. Liu, M. Ott, N. Goyal, et al., "RoBERTa: A Robustly Optimized BERT Pretraining Approach," 2019, arXiv Preprint arXiv:1907.11692.
29. G. Salton and C. Buckley, "Term-Weighting Approaches in Automatic Text Retrieval," *Information Processing & Management* 24, no. 5 (1988): 513–523.
30. N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings Using Siamese BERT-Networks," in *Proceeding Conference on Empirical Methods in Natural Language Processing and the International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)* (ACL Anthology, 2019), 3982–3992.
31. G. Ke, Q. Meng, T. Finley, et al., "LightGBM: A Highly Efficient Gradient Boosting Decision Tree," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30 (NeurIPS, 2017), 3146–3154.